# VINCRISTINE

FATAL IF GIVEN INTRATHECALLY

Vesicant; neurotoxic

**CHEMICAL NAME** Vinca alkaloid from *Catharanthus roseus* ($C_{46}H_{56}N_4O_{10}$, sulfate)
**CLASS** Vinca alkaloid (microtubule inhibitor) cytotoxic
**TRADE NAMES** Oncovin · Vincasar · Vincristine sulfate injection

## THERAPEUTIC USES / INDICATIONS

- Acute lymphoblastic leukaemia (induction).
- Lymphomas (CHOP, R-CHOP), Hodgkin's.
- Paediatric tumours: Wilms', neuroblastoma, rhabdomyosarcoma.
- Small cell lung cancer; ITP (refractory).

## ADMINISTRATION

- **IV ONLY** — diluted in a 50 mL minibag (not syringe) to prevent intrathecal error; label "For intravenous use only — fatal if given by other routes".
- Give via free-flowing drip; vesicant (extravasation → necrosis; warm compress + hyaluronidase).
- Never in the same session/area as intrathecal chemotherapy (separate time and place).

## DOSAGES

- **Adult:** 1.4 mg/m² IV (dose usually capped at 2 mg) weekly or per cycle.
- **Children:** 1.5–2 mg/m² (max 2 mg); <10 kg 0.05 mg/kg.
- Reduce in hepatic impairment (bilirubin >3 mg/dL: 50%).

## PHARMACOKINETICS

- Rapidly distributed; metabolised by CYP3A4.
- Excreted mainly in bile. **t½:** terminal ~85 h.

## PHARMACODYNAMICS

- Binds **tubulin** → inhibits microtubule assembly → metaphase arrest (M phase); also impairs axonal transport (neurotoxicity).

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Constipation (autonomic neuropathy), abdominal pain.
- Peripheral neuropathy: tingling, loss of reflexes, foot drop.
- Alopecia; jaw pain.

### ADVERSE EFFECTS (SERIOUS)

- Intrathecal administration — ascending paralysis and death.
- Paralytic ileus.
- Severe neuropathy, cranial nerve palsies (ptosis, hoarseness), SIADH.
- Extravasation necrosis (myelosuppression is minimal).

## CONTRAINDICATIONS & CAUTIONS

- Intrathecal route (absolute).
- Demyelinating Charcot–Marie–Tooth disease.
- Severe pre-existing neuropathy.
- Pregnancy.

### KEY INTERACTIONS

- Itraconazole, posaconazole, voriconazole, ketoconazole: severe neurotoxicity (CYP3A4); avoid.
- **Phenytoin:** ↓ phenytoin levels.
- **Live vaccines:** contraindicated.

## NURSING MANAGEMENT

- Independent checks of route and label; minibag only; never alongside intrathecal drugs.
- Monitor bowel habit daily; prophylactic laxatives.
- Assess neuropathy (reflexes, fine motor, gait, jaw pain) before each dose.
- Check cannula patency; extravasation management.
- Monitor $Na^+$ (SIADH).

## PATIENT EDUCATION

- Take laxatives as prescribed; report no bowel movement for 2–3 days or abdominal pain.
- Report numbness, tingling, weakness, trouble walking or doing buttons, drooping eyelid, hoarse voice.
- Report pain at the drip site during infusion.

**KEY POINTS**

- Vinca alkaloid — inhibits microtubule assembly.
- Neurotoxicity and constipation (minimal myelosuppression).
- IV only (minibag); intrathecal is fatal.
- Dose capped at 2 mg; avoid azoles.